Data Communication & Computer Networks (CN)

Topic 2 of 7 : Network Models (OSI & TCP/IP)

0 MCQs — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** MCQs

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic2_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

P1. Fiber Optic Cables and Coaxial Cables function at which layer

(A) Data Link Layer
(B) Physical Layer
(C) Network Layer
(D) Transport Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P2. The term "Baud Rate" is associated with which layer

(A) Network Layer
(B) Data Link Layer
(C) Physical Layer
(D) Application Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P3. Which layer creates the "Collision Domain"? / "Collision Domain" layer create

(A) Physical Layer (Hubs)
(B) Data Link Layer (Switches/Bridges)
(C) Network Layer
(D) Transport Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P4. Assertion

(A) : The Data Link Layer divides the stream of bits received from the Network Layer into manageable data units called Frames. (A): Reason (R): Framing is necessary to separate one message from another and to provide synchronization. (R): (A) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is NOT the correct explanation.
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P5. Determine the OSI layer responsible for dialogue control (half-duplex or full-duplex communication). OSI model dialogue control (half-duplex full-duplex communication) layer

(A) Session Layer
(B) Transport Layer
(C) Network Layer
(D) Data Link Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P6. Which layer translates data from a format used by the application into a common format (e.g., ASCII to EBCDIC)? layer application format common format (ASCII to EBCDIC) translate

(A) Application Layer
(B) Presentation Layer
(C) Session Layer
(D) Transport Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P7. Identify the correct order of data encapsulation from Layer 7 to Layer 1:

(A) Data → Segment → Packet → Frame → Bits
(B) Data → Packet → Segment → Frame → Bits
(C) Bits → Frame → Packet → Segment → Data
(D) Data → Frame → Packet → Segment → Bits

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P8. The Physical layer deals with: **(A)** Logical addressing **(B)** Media, signals and binary transmission **(C)** Error correction **(D)** Segmentation [Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]	⚠ auto-ingested — answer needs verification
P9. The sublayer of Physical layer which directly interfaces with MAC sublayer is called: MAC **(A)** Physical signalling sublayer **(B)** Physical data sublayer **(C)** Physical address sublayer **(D)** Physical transport sublayer [Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]	⚠ auto-ingested — answer needs verification
P10. The Physical layer is responsible for which of the following operations **(A)** Line coding **(B)** Channel coding **(C)** Modulation **(D)** All of the above [Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]	⚠ auto-ingested — answer needs verification
P11. Data Link Layer is divided into: **(A)** LLC and MAC **(B)** Physical and Network **(C)** Transport and Session **(D)** Application and Presentation [Source: BCI Subject Test-8.pdf | auto-classified: Topic 2 / OSI Model Basics]	⚠ auto-ingested — answer needs verification
P12. Match the following Security Protocols with their corresponding OSI Layers: # % + - 2 OSI 9 : List–I (Protocol) List–II (Layer) **(A)** IPSec (i) Application Layer **(B)** SSL / TLS (ii) Network Layer **(C)** PGP (Pretty Good Privacy) (iii) Transport / Presentation Layer **(D)** WPA2 (iv) Data Link Layer (Wi-Fi) Choose the correct answer from the codes given below: (A) a–ii, b–iii, c–i, d–iv (B) a–ii, b–i, c–iii, d–iv (C) a–iii, b–ii, c–i, d–iv (D) a–i, b–ii, c–iii, d–iv [Source: BCI Subject Test-9.pdf | auto-classified: Topic 2 / OSI Model Basics]	⚠ auto-ingested — answer needs verification

P13. Security Mechanisms & OSI Layers + X OSI B List–I (Mechanism / X)
(A) Firewall ( )
(B) Digital Signature (W +)
(C) MAC Filtering (MAC YZ)
(D) Port-based Filtering (9 Z YZ) List–II (Layer /) (i) Network Layer / 9 (ii) Application Layer / %\ (iii) Data Link Layer (iv) Transport Layer / S 9 Codes / : (A) a–i, b–ii, c–iii, d–iv (B) a–ii, b–i, c–iii, d–iv (C) a–iv, b–ii, c–i, d–iii (D) a–i, b–iii, c–ii, d–iv

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P14. Which statement best describes the difference between LLC and MAC sublayers? LLC MAC B -
(A) LLC handles physical addressing, MAC handles flow control
(B) LLC provides service interface to Network layer, MAC controls access to medium
(C) LLC works only in wireless networks, MAC only in wired networks
(D) LLC performs framing, MAC performs routing

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P15. The primary difference between active and passive attacks is _____. Active Passive + T
(A) Active attacks change data; passive do not.
(B) Passive attacks destroy hardware.
(C) Both are same.
(D) Passive attacks are legal.

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P16. Which OSI layer is responsible for logical addressing? - OSI 3
(A) Data Link
(B) Network
(C) Transport
(D) Application

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P17. Which layer performs segmentation
(A) Data link
(B) Transport
(C) Network
(D) Session

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P18. What is the PDU of Data Link layer? PDU 7
(A) Packet
(B) Frame
(C) Segment
(D) Bit

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P19. OSI model was developed by OSI 9
(A) IEEE
(B) ISO
(C) IETF
(D) ITU
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P20. Which layer is responsible for encryption? =>? @ B
(A) Session
(B) Presentation
(C) Application
(D) Transport
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P21. Which layer provides end-to-end communication
(A) Network
(B) Transport
(C) Session
(D) Physical
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P22. Encapsulation happens from: 0 7 ?**
(A) Top to bottom
(B) Bottom to top
(C) Random
(D) Parallel
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P23. Consider the following statements: S1: Network layer handles both logical addressing and routing. S2: Transport layer handles segmentation and reassembly. 9 : 7: S1: A E S2: E A 7
(A) Only S1
(B) Only S2
(C) Both S1 and S2
(D) Neither S1 nor S2
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P24. In OSI model, which of the following layers participate in “end-to-end” delivery? OSI 7 “ - - ” 7 7 O?** (1) Network (2) Transport (3) Session (4) Data link
(A) 1,2 only
(B) 2 only
(C) 1,2,3
(D) 2,3,4
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P25. A frame is received at Data Link layer with incorrect FCS. What will happen? FCS P R ?**

(A) Frame discarded
(B) Frame corrected
(C) Sent to network layer
(D) Re-transmitted automatically

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P26. In TCP/IP model, the responsibilities of Session layer are part of: TCP/IP 7 Session S T V O?**

(A) Internet layer
(B) Network Access layer
(C) Application layer
(D) Transport layer

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / TCP/IP Model]

⚠ auto-ingested — answer needs verification

P27. Which layer performs “service point addressing”? " A E " ?**

(A) Network
(B) Data Link
(C) Transport
(D) Session

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P28. Which statement is TRUE regarding encapsulation? 0 7 ?**

(A) Header is removed at sender
(B) Trailer added at Network layer
(C) Header size always remains constant
(D) Each layer adds its own header

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⚠ auto-ingested — answer needs verification

P29. Which OSI layer ensures interoperability between different data formats? \ ^ a ?**

(A) Presentation
(B) Application
(C) Session
(D) Network

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P30. Physical addressing is added at: ?**

(A) Network
(B) Transport
(C) Data Link
(D) Session

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⚠ auto-ingested — answer needs verification

P31. OSI model has how many layers? OSI 7 7 O?**
(A) 5
(B) 6
(C) 7
(D) 8
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P32. Which OSI layer handles message boundary? (boundary)
(A) Presentation
(B) Session
(C) Transport
(D) Network
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P33. “Transparent transmission” is the function of which layer?" E 7 E " A ?**
(A) Data link
(B) Network
(C) Session
(D) Transport
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P34. Which layer supports “checkpointing”? " u " 7 ? **
(A) Transport
(B) Application
(C) Session
(D) Data link
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P35. Which OSI layer provides “naming conventions”? - " v7 " W ?**
(A) Transport
(B) Session
(C) Presentation
(D) Application
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P36. Which layer handles “hop-to-hop” flow control? " - - w E " 7 ?**
(A) Network
(B) Data Link
(C) Transport
(D) Physical
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P37. Which OSI layer decides the type of compression? x W A 7 ?**
(A) Application
(B) Presentation
(C) Transport
(D) Data link
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P38. Which layer is responsible for “peer- to-peer communication”? " - - " 7 ?**
(A) Session
(B) Network
(C) Transport
(D) Data Link
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P39. Which layer handles end-to-end delivery of messages
(A) Network Layer
(B) Transport Layer
(C) Data Link Layer
(D) Application Layer
[Source: CN Topic Test 6.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P40. What is the role of the TTL field in an IP packet? IP TTL
(A) Authentication
(B) Packet size control
(C) Packet lifespan
(D) Encryption
[Source: CN Topic Test 6.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P41. Which field in IPv4 header indicates the priority of the packet?IPv4
(A) Protocol
(B) TTL
(C) Type of Service (ToS)
(D) Flags
[Source: CN Topic Test 6.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P42. Which OSI layer is responsible for encryption and compression? OSI
(A) Presentation Layer
(B) Session Layer
(C) Transport Layer
(D) Network Layer
[Source: CN Topic Test 6.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P43. IPv4 and IPv6 are at which OSI layer?IPv4 IPv6 OSI
(A) Data link layer
(B) Network layer
(C) Transport layer
(D) application layer
[Source: CN Topic Test 6.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P44. Which of the following is not part of TCP/IP model? TCP/IP
(A) Application
(B) Transport
(C) Data link
(D) Session Which tool is used for packet sniffing? (a) Putty (b) Wireshark (c) File Zilla (d) nslookup
[Source: CN Topic Test 6.pdf | auto-classified: Topic 2 / TCP/IP Model]

⚠ auto-ingested — answer needs verification

P45. Which layer is responsible for reliable delivery in OSI?OSI
(A) Session
(B) Transport
(C) Network
(D) Application
[Source: CN Topic Test 5.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P46. What is jitter in networking
(A) Delay in propagation
(B) Variation in packet arrival time
(C) Packet loss
(D) Transmission error
[Source: CN Topic Test 5.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P47. Transmission data rate is decided by
(A) network layer
(B) physical layer
(C) data link layer
(D) transport layer
[Source: CN Topic Test-01.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P48. TCP/IP model does not have _____ layer but OSI model have this layer. _____
(A) session layer
(B) presentation layer
(C) application layer
(D) both (a) and (b)
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P49. OSI stands for
(A) open system interconnection
(B) operating system interface
(C) optical service implementation
(D) none of the mentioned
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P50. TCP/IP model was developed _____ the OSI model.
(A) prior to
(B) after
(C) simultaneous to
(D) none of the mentioned
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P51. Which layer is responsible for process to process delivery
(A) network layer
(B) transport layer
(C) session layer
(D) data link layer
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P52. Which layer provides the services to user
(A) application layer
(B) session layer
(C) presentation layer
(D) physical layer
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P53. In the layer hierarchy as the data packet moves from the upper to the lower layers, headers are
(A) Added
(B) Removed
(C) Rearranged
(D) Modified
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P54. In OSI model, when data is sent from device A to device B, the 5th layer to recieve data at B is.. OSI A B
(A) Application layer
(B) Transport layer
(C) Link layer
(D) Session layer
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P55. The TCP/IP _____ layer is equivalent to the combined session, presentation, and application layers of the OSI model. TCP/IP _____
(A) application
(B) network
(C) data link
(D) physical
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / TCP/IP Model]

⚠ auto-ingested — answer needs verification

P56. The _____ layer is responsible for the source-to-destination delivery of a packet across multiple network links. _____
(A) transport
(B) network
(C) data link
(D) physical
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P57. The _____ layer is responsible for delivering data units from one station to the next without errors. _____
(A) transport
(B) network
(C) data link
(D) physical
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P58. The session, presentation, and application layers are the ____ support layers. ____
(A) user
(B) network
(C) both (a) and (b)
(D) neither (a) nor (b)
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P59. Why was the OSI model developed? OSI
(A) Manufacturers disliked the TCP/IP protocol suite
(B) The rate of data transfer was increasing exponentially
(C) Standards were needed to allow any two systems to communicate
(D) None of the above
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P60. Which of the following layer is not network support layer
(A) Transport Layer
(B) Network Layers
(C) Data link Layer
(D) Physical Layer
[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P61. Which layer helps to understand data representation in one form on a host to other host in their native representation

(A) Application Layer
(B) Presentation Layer
(C) Session Layer
(D) Transport Layer

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⚠ auto-ingested — answer needs verification

P62. From which layer does the data link layer collect packets and encapsulate them into frames for transmission

(A) Transport layer
(B) Application layer
(C) Network layer
(D) Physical layer

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⚠ auto-ingested — answer needs verification

P63. A Server at a company called ATech is using TCP/IP protocol to connect to the Network & is running multiple applications. At which layer of OSI Model is this Server operating: ATech TCP/IP OSI

(A) Application Layer
(B) Physical Layer
(C) Network Layer
(D) Transport Layer

[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P64. Layer that is used to deal with mechanical and electrical specifications is:

(A) Physical Layer
(B) Data Link Layer
(C) Network Layer
(D) Transport Layer

[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P65. _____ Layer is responsible for Framing, Error Detection and Physical Addressing. _____

(A) Network
(B) Transport
(C) Application
(D) Data Link _____ layer defines how data is formatted, presented, encoded, and converted for use on the network. _____ (a) Transport (b) Presentation (c) Application (d) Physical

[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P66. 11, IEEE 802.15, IEEE 802.16 (b) POTS, PDH, SONET, SDH, PON, OTN, DSL, ITU-T V series, ITU-T G.hn PHY (c) I.430, I.431, EIA/TIA-232, EIA/TIA-449 (d) All of the above 41. Who developed standards for the OSI reference model? OSI

(A) ANSI - American National Standards Institute
(B) ISO - International Standards Organization
(C) IEEE - Institute of Electrical and Electronics Engineers
(D) ACM - Association for Computing Machinery

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⚠ auto-ingested — answer needs verification

P67. Network support layers are

(A) 1, 2, 3
(B) 4,5
(C) 5,6,7
(D) 4

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⚠ auto-ingested — answer needs verification

P68. Bridge works in which layer of the OSI model? OSI

(A) Application layer
(B) Transport layer
(C) Network layer
(D) Data link layer

[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P69. A computer network permits sharing of

(A) Resources
(B) Information
(C) Both a & b
(D) None of these Which layers of the OSI model are host to host layers? OSI (a) transport,session,Presentation, Application (b)Network,Transport,Session, Presentation (d) Datalink, Network, Transport, Session (e) Physical,DataLink,Network,Trans port

[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P70. Which layer connects the network support layers and user support layers

(A) Transport layer
(B) Network layer
(C) Data link layer
(D) Session layer

[Source: CN Topic Test-02.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P71. Assume that Source S and Destination D are connected through an intermediate router R. How many times a packet has to visit the network layer and data link layer during a transmission from S to D? S D R S

(A) Network layer - 4 times, Data link layer – 4 times
(B) Network layer - 4 times, Data link layer - 6 times
(C) Network layer - 2 times, Data link layer - 4 times
(D) Network layer - 3 times, Data link layer - 4 times

[Source: CN Topic Test-03.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P72. Repeaters function in

(A) Physical layer
(B) Data link layer
(C) Network layer
(D) Both (A) and (B)

[Source: CN Topic Test-03.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P73. Consider ISO-OSI network architecture reference model. Session layer of this model offer Dialog control, token management and_____ as services. ISO-OSI _____

(A) Synchronization
(B) Asynchronization
(C) Errors
(D) Flow control

[Source: CN Topic Test-03.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P74. Simple network management protocol(SNMP) is a protocol that runs on: ( )

(A) Application layer
(B) Transport layer
(C) Network layer
(D) Data link layer

[Source: CN Topic Test-03.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P75. Which organization is responsible for managing IP address allocation globally

(A) IEEE
(B) ICANN
(C) IETF
(D) ISO

[Source: CN Topic Test-03.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P76. Which physical layer technique allows multiple users to share the same bandwidth simultaneously

(A) TDMA
(B) FDMA
(C) CDMA
(D) ARQ

[Source: CN Topic Test-04.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P77. The reliable data transfer is implemented by

(A) Network layer
(B) Physical layer
(C) Presentation layer
(D) Transport layer

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P78. In IPv4, packet fragmentation and reassembly is handled by

(A) using the More Fragments and Fragment Offset Fields
(B) using the MAC Header
(C) using the Header Checksum
(D) encapsulating packets in larger data frames

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P79. How many address types are used from sender to receiver in OSI communication? OSI 7 W W Y A E W O?**

(A) 2
(B) 3
(C) 4
(D) 5

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 1 / DC Components & Protocol Basics]

⚠ auto-ingested — answer needs verification

P80. The number of layers in Internet protocol stack

(A) 5
(B) 7
(C) 6
(D) 3

[Source: CN Topic Test-02.pdf | auto-classified: Topic 1 / DC Components & Protocol Basics]

⚠ auto-ingested — answer needs verification

P81. Application layer protocol defines:

(A) types of messages exchanged
(B) rules for when and how processes send and respond to messages
(C) message format, syntax and semantics
(D) all of above

[Source: CN Topic Test-03.pdf | auto-classified: Topic 1 / DC Components & Protocol Basics]

⚠ auto-ingested — answer needs verification

P82. The physical layer is responsible for -

(A) Bit-by-bit delivery
(B) Port to port delivery
(C) Process to process delivery
(D) Application to application delivery

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 1 / Characteristics & Criteria]

⚠ auto-ingested — answer needs verification